

Tharun Kunamalla

Software Engineer | Backend Development, Java, Cloud, DSA

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EDUCATION

Indian Institute of Information Technology Kottayam

Kottayam, Kerala

Bachelor of Technology in Computer Science and Engineering

Aug 2023 – May 2027

- **Relevant Coursework:** Data Structures, Algorithms, Computer Programming, Databases, Machine Learning, Artificial Intelligence, Computer Networks, Operating Systems, Object-Oriented Programming

TECHNICAL SKILLS

Programming Languages: Java, Python, JavaScript, TypeScript, C++, SQL

Backend Technologies: Node.js, Express.js, Spring Boot, FastAPI, REST APIs, Microservices, Socket.IO, JWT, Apache Kafka

Frontend Technologies: React.js, HTML, CSS, Tailwind CSS, Leaflet

Databases: MongoDB, MySQL

Cloud & DevOps: Google Cloud Platform (GCP), AWS, Docker, Nginx, GitHub Actions (CI/CD)

Testing & Tools: JUnit, Vitest, Git, GitHub, Linux, Postman, Jupyter Notebook

Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

PROFESSIONAL EXPERIENCE

AI/ML Intern

Jun 2025 – Aug 2025

Labmentix

Bengaluru, India

- Built XGBoost, Random Forest, and Decision Tree models, achieving 87% accuracy through feature engineering, EDA on 10K+ records, and GridSearchCV hyperparameter tuning.

RESEARCH EXPERIENCE

Research Assistant

May 2026 – July 2026

National Institute of Technology Warangal (Under Prof. Venkateshwara sir)

Warangal, India

- Contributed to the UniRecSys group package recommendation system by extending the DREX framework to SGDREX using GRU, BERT, and collaborative filtering.
- Evaluated group preference aggregation models on real-world datasets to improve personalized recommendation quality.

PROJECTS

Instant Route Guide – Graph Search Platform | *Java, Spring Boot, React, Node.js, Kafka, Docker* | [Live](#) | [GitHub](#)

- Built a **microservices-based routing platform** using React/Vite, Spring Boot, Node.js, Nginx, and Docker Compose to compare **BFS, Dijkstra, A*, Bidirectional Dijkstra, and Bidirectional A*** on OpenStreetMap road networks with real-time **Leaflet** visualization and geocoding APIs.
- Implemented **in-memory caching** for efficient path computation and an **Apache Kafka telemetry pipeline** for search-metric streaming, aggregation, and consumer retries, with **Vitest/JUnit** testing and **GitHub Actions CI/CD**.

CodeSync – Real-Time Collaborative Code Editor | *MERN, Socket.IO, Monaco Editor, JWT, Docker* | [Live](#) | [GitHub](#)

- Built a real-time collaborative code editor using the MERN stack, Socket.IO, and Monaco Editor with concurrent editing, JWT authentication, and multi-language code execution.
- Implemented Socket.IO-based communication, JWT authentication, and conflict resolution to ensure low-latency collaborative editing.

Wavvy – Real-Time Watch Party Platform | *MERN, Socket.IO, WebRTC, MongoDB, Docker* | [Live](#) | [GitHub](#)

- Built a **real-time watch party platform** using React, Node.js, Express, Socket.IO, and MongoDB supporting synchronized video playback, room management, live chat, and shared playlists.
- Implemented **WebRTC peer-to-peer video calling** with Socket.IO signaling, including offer/answer exchange and ICE candidate handling for in-room communication.

LEADERSHIP, OPEN SOURCE & ACHIEVEMENTS

Google Summer of Code (GSoC) 2026 Contributor | Open Source

- Contributed to **OmegaUp** and **Sugar Labs – Music Blocks** through feature development, bug fixes, and GitHub pull requests.

GirlScript Summer of Code (GSSoC'25) Contributor

- Completed **20+ open-source contributions**, collaborating with maintainers through issue resolution, code reviews, feature development, and GitHub pull requests.

Sub-Lead, Robotics Club | BetaLabs, IIIT Kottayam

Sep 2024 – Oct 2025

Solved 500+ Data Structures and Algorithms problems on LeetCode using Java.

CERTIFICATIONS

Microsoft Azure AI | Labmentix AI/ML Training | AWS Cloud Essentials | Postman API Expert